Asset Performance:

Hea 03 Internal and external lighting levels

6 Credits

0

No Minimum Standard

Aim

To ensure appropriate lighting is provided to enable asset users to perform visual tasks efficiently and accurately.

Question

Do internal and external lighting levels meet best practice illuminance (lux) levels in occupied space?

Credits	Answer	Select either C or D, Answer E can be selected independently
0	A.	Question not answered
0	B.	No
2	C.	Yes, ≥ 50% of occupied space meets best practice illuminance (lux) levels for internal lighting
4	D.	Yes, ≥ 80% of occupied space meets best practice illuminance (lux) levels for internal lighting
2	E.	Yes, ≥ 80% of relevant external space meets best practice illuminance (lux) levels for external lighting

Assessment criteria

Criterion	Assessment criteria	Applicable Answer
1.	**Filtering** The external lighting credits can be filtered out in the following circumstances: a) Where no external light fittings are installed (either separate from or mounted on the external asset's façade or roof) and external lighting is not required for safety or task performance reasons. OR b) External spaces are not within scope of the assessment.	E
2.	Credits are awarded based on the proportion of internal or external space that meets the relevant best practice illuminance levels (average illuminance and illuminance uniformity) for task areas in each space. For the	C – E

Criterion	Assessment criteria	Applicable Answer
	purposes of this assessment issue, the following standards are considered to contain best practice illuminance (lux) levels: • EN 12464-1:2011 Light and lighting – Lighting of work places - Part 1: Indoor work places • EN 12464-2:2014 Light and lighting – Lighting of work places - Part 2: Outdoor work places • Society of Light and Lighting (SLL), Code for Lighting, 2012 • Illuminating Engineering Society (IES), The Lighting Handbook 10th Edition, 2011. Alternatively, where local standards have similar requirements to any of the above standards, the local standard may be used to demonstrate compliance with this criterion.	
3.	Illuminance levels are measured in occupied space by a suitably qualified person in accordance with the procedure(s) in the relevant best practice standard (criterion 2). Where the best practice standard does not stipulate procedures for measuring illuminance, the procedure in the methodology section must be followed. If during measurement light fittings need to be added or removed or covered in an occupied space to achieve the best practice illuminance levels, the occupied space would not comply with the criteria.	C – E
4.	External spaces within the scope of the external lighting requirements include, but are not limited to, the following: • Walkways exclusively for pedestrians • Traffic areas for slowly moving vehicles (maximum 10 km/h), e.g. bicycles, trucks and excavators • Regular vehicle traffic (maximum 40 km/h) • Vehicle turning, loading and unloading points • Parking areas.	E

Methodology

Procedure for measuring illuminance

The suitably qualified person must confirm that they have used either the illuminance verification procedure(s) stipulated within the relevant best practice standard (criterion 2) or where the best practice standard does not stipulate procedures for measuring illuminance, the following procedure:

- Illuminance measurements should be undertaken using calibrated illuminance meters with a photocell that is both colour/spectrally and cosine corrected. The deviation of the calibration values as per the calibration certificate should be applied when analysing the measurement results. Additionally, it should be checked that the ambient temperature during the measurements does not depart markedly from the ambient temperature for which the photocell had been calibrated (typically, this is 25°C); if this is the case, then the measured values should also be corrected for ambient temperature.

- Illuminance meters with an integrated photocell should be checked for the correct levelling on the measurement surface. In the case of illuminance meters that have a detachable photocell, the photocell should be secured inside a specially designed holder and the holder should be checked for the correct levelling on the measurement surface.
- The mains voltage supply in the test areas should be stable during measurements.
- It should be checked that there are no unusual temperature differences between different zones of the test areas. The artificial lighting should also be kept on long enough so that its light output stabilises (e.g. at least 20 minutes for fluorescent, discharge or LED lighting).
- Indoor illuminance measurements should be carried out so that daylight penetration is avoided completely whilst taking the readings. In spaces with windows, illuminance measurements would ideally be undertaken after dusk; if blinds are fitted, these should be closed in order to minimise any potential stray light from buildings nearby and external lighting. Measurements of outdoor illuminance should be carried out when it is fully dark and in good weather conditions.
- Items within the measurement areas should be kept to a level that would be expected during normal use. Additionally, the suitably qualified person should be far enough away from the photocell, ideally below its height if possible, to avoid casting shadows or preventing light from being reflected onto the photocell.
- Illuminance measurements are normally required to assess average illuminance or illuminance uniformity on the working plane or other task areas that can be horizontal, vertical or inclined. In existing buildings, task areas, e.g. fixed desks, inclined planes of industrial equipment or vertical storage rack faces, are generally known. In an open task area, a grid of measurement points should be set out. The distance between the measurement points p is calculated using the formula:

$$p = 0.2 \ x \ 5^{(log_{10} d)}$$

Where d is the length of the longer dimension of the area being measured, and thus the number of points in the relevant dimension is given by the nearest whole number of d/p. The resulting spacing between the grid points is used to calculate the nearest whole number of grid points in the other dimension. Once illuminance is measured at each grid point, average illuminance can be determined as the arithmetic mean of the measured values. Illuminance uniformity can be calculated as the ratio of the minimum illuminance (as measured) to the average illuminance (as calculated).

- For horizontal task planes, a perimeter zone of 0.5m from the walls can be excluded from the measurement area except when a task area is in or extends into this perimeter zone.
- Alternatively, spot readings of illuminance may be taken at all locations where people are working, for example on each desk in an open plan office. This is acceptable practice where measurement grids cannot be defined due to the presence of obstructions in the measured areas. An occupied space is deemed compliant if both the average illuminance and the average uniformity ratio (given by minimum illuminance divided by average illuminance) are within standard recommendations.
- For spaces with more than one type of task area, each with different best practice illuminance levels, compliance is achieved if the average illuminance and the average uniformity ratio for all task area types are within the best practice recommendations.
- For assets with identical lighting systems and floor layouts across multiple areas or floors, illuminance levels in a representative sample of areas or floors may be measured, if in the professional opinion of the suitably qualified person, these are likely to give measurements that reflect the performance of the lighting in all relevant spaces within the asset. Illuminance measurements must be taken at least every

five years and after any changes to lighting systems. Where the asset is less than five years old and illuminance levels were assessed as part of the design or construction of the asset that demonstrate compliance with the criteria in this assessment issue, then the results of these studies may be used as evidence (assuming there have been no changes to the lighting systems in the intervening period).

Calculating percentage of internal and external space with compliant illuminance levels

For internal lighting, this is based on net internal floor area as follows:

Percentage of occupied space with compliant illuminance levels for internal lighting =

$$\left(\frac{\begin{gathered}\textit{Net internal area of occupied space with compliant illuminance}\\ \textit{levels for internal lighting}\end{gathered}}{\begin{gathered}\textit{Total net internal area of all occupied space requiring compliant illuminance}\\ \textit{levels for internal lighting}\end{gathered}}\right)\times 100$$

For example, for an asset with a total net internal area of all occupied space of 1,000 m^2, at least 500 m^2 of net internal floor area of the occupied space must meet the best practice illuminance levels for internal lighting to achieve any credits.

For external lighting, this is based on the area of relevant external space as follows (where any external space does not fall within the scope of criterion 4, then this should be excluded from this calculation):

Percentage of external space with compliant illuminance levels for external lighting =

$$\left(\frac{\begin{gathered}\textit{Area of relevant external space with compliant illuminance}\\ \textit{levels for external lighting}\end{gathered}}{\begin{gathered}\textit{Total area of all relevant external space requiring compliant illuminance}\\ \textit{levels for external lighting}\end{gathered}}\right)\times 100$$

For example, for an asset with a total area for all relevant external spaces of 1,000 m^2, at least 800 m^2 of the relevant external spaces must meet the best practice illuminance levels for external lighting to achieve the credits.

Evidence

Criteria	Evidence requirement
-	The evidence below is not exhaustive, please also refer to the 'BREEAM evidential requirements' section in the scope of the Guidance for appropriate evidence types which can be used to demonstrate compliance.
2 - 4	Documentation confirming the illuminance levels in occupied space meet the best practice levels and have been measured in accordance with the 'Procedure for measuring illuminance' (see Methodology).

Definitions

Illuminance:

The amount of light falling on a surface per unit area, measured in lux.

Occupied space:

A room or space within the asset that is likely to be continuously occupied for 30 minutes or more per day by an asset user.

Suitably qualified person:

Someone that has demonstrable experience or training in undertaking lighting measurements in internal and external spaces.